

DANMARKS TEKNISKE UNIVERSITET



MODELING CO₂ CONCENTRATIONS USING HIDDEN
MARKOV MODELS

2026

SPECIAL COURSE IN STATISTICAL MODELLING

Mads Holme Håkonsson
s234936

Supervisor
Jan Klobbenborg Møller

29. maj 2026

AI Declaration

AI tools has been used for formatting LaTeX figures, tables and equations and grammar checking in the report. No AI has been used for for writing the math or the academic content of the report or the code. No Agentic AI tools has been used for wrriting the src code or the report. Agentic AI tools has been used for writing some of testing files in the tests directory or refactoring code as renaming files, variables or functions.

Indhold

1	Introduction	3
----------	---------------------	----------

1 Introduction

This report is written for the purpose of given a concise outline of the projects achivements, design choices, HMM Model results and interpretaiton. The report is written for future students to be able to easilyff understssand the projectdd. dd. rw. Teldo. f. . Te thello. f. . . ssss. ss. ssss.